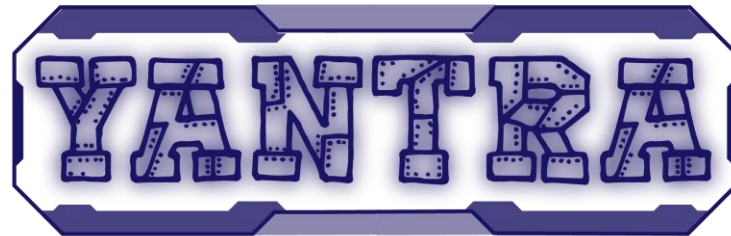


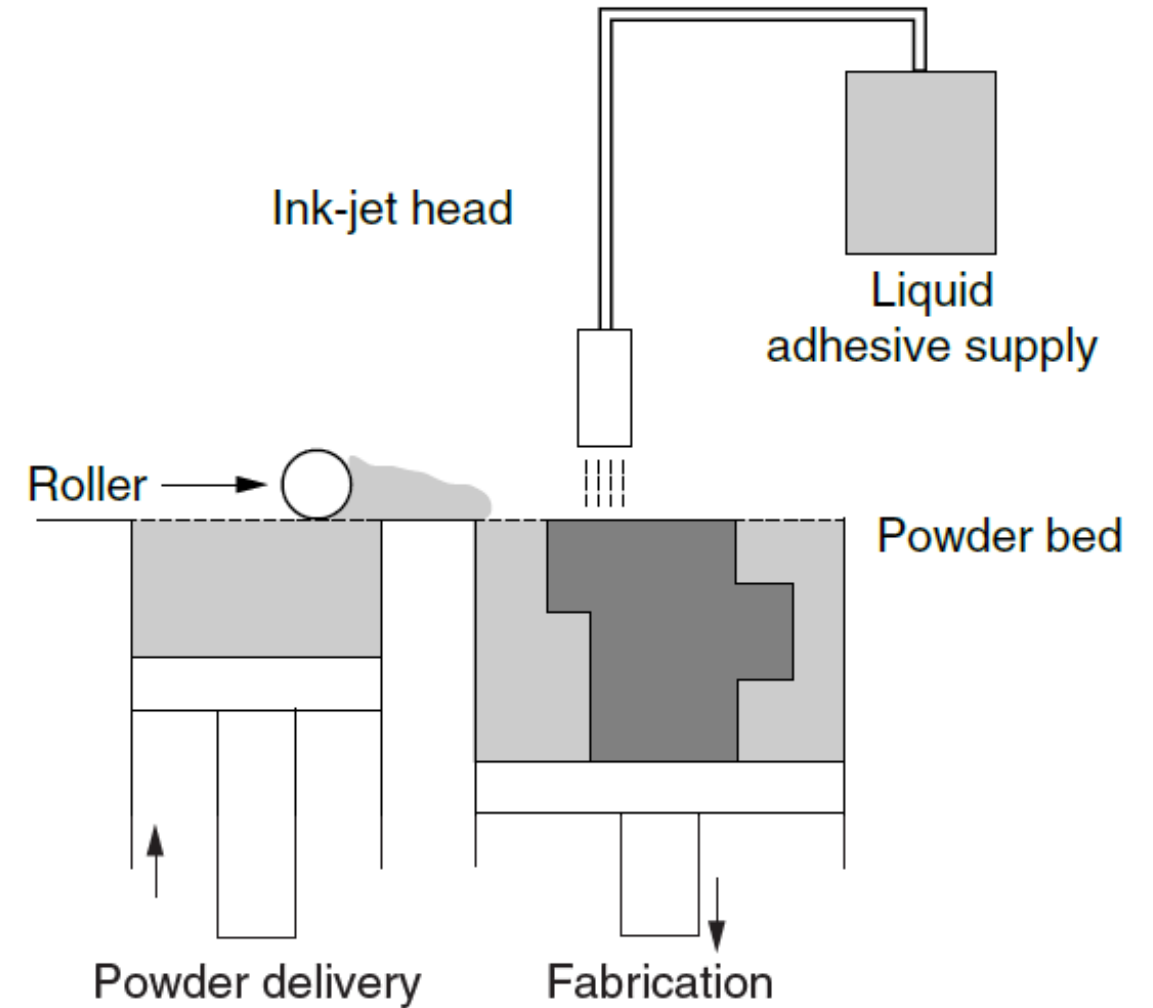
Three-dimensional printing

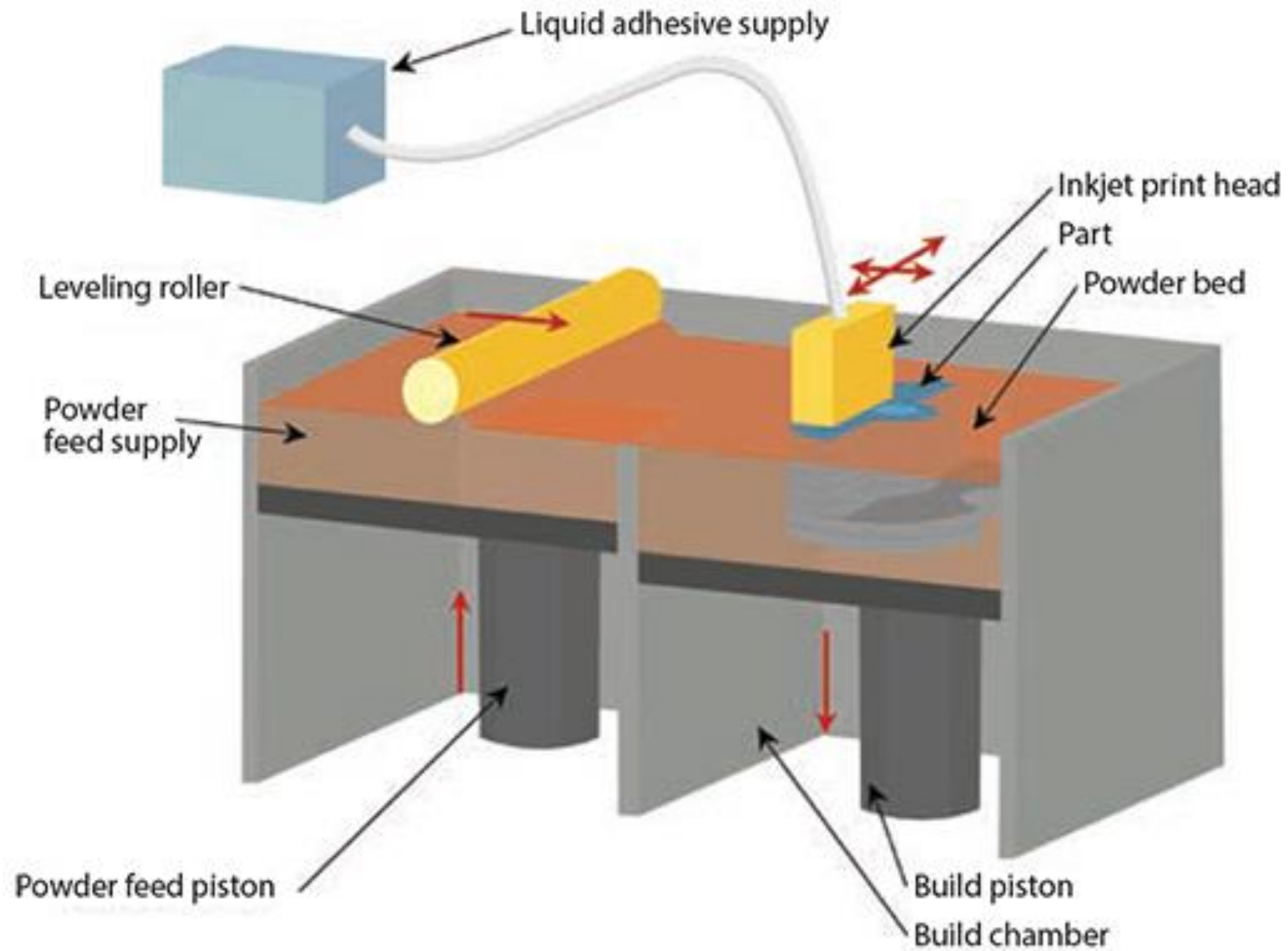


fastest and the least expensive way to create large appearance prototypes.

The system also offers a variety of **finishing options** including epoxy infiltration, sanding, painting and plating.

The **option of color** gives the user added information and aesthetics through the ability to incorporate color directly into the part as it is **being printed**

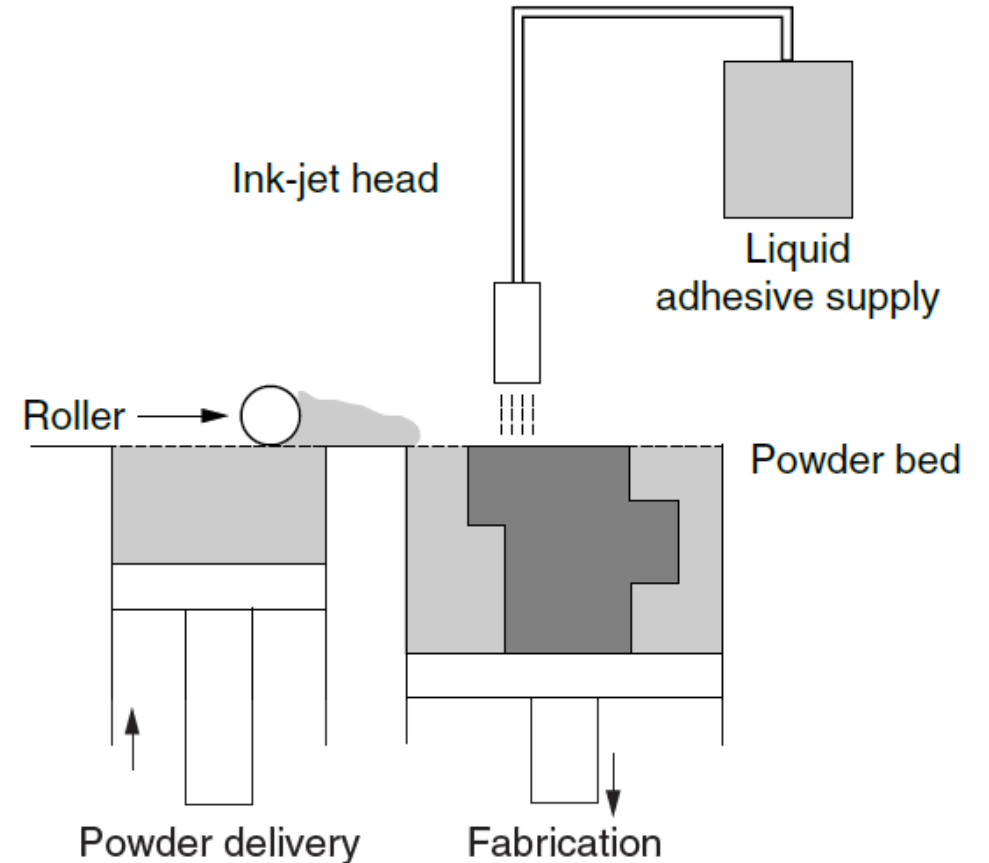




Process

Powder material is deposited in successive layers, as in SLS. The process starts by **depositing a layer of powder** object material at the top of a fabrication chamber.

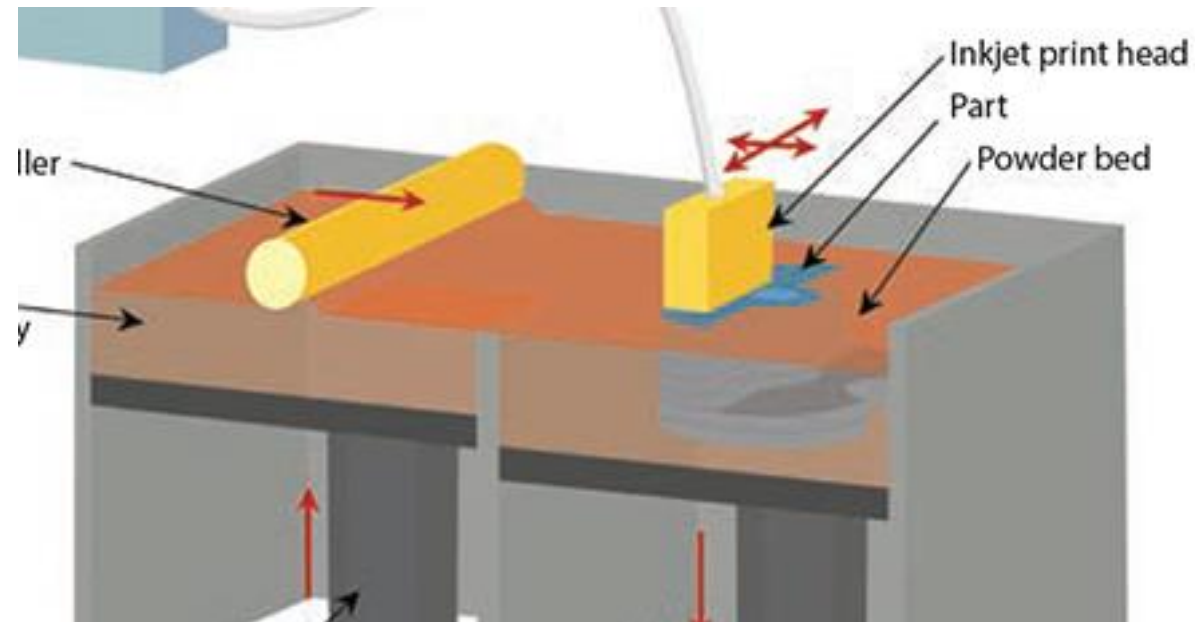
To accomplish this, a measured quantity of powder is first dispensed from a similar supply chamber by a piston moving upward incrementally.



The **roller** then **distributes** and **compresses** the powder at the top of the fabrication chamber.

The multichannel jetting head subsequently deposits a **liquid adhesive** in a two-dimensional pattern onto the layer of the powder,

which becomes bonded in the areas where the adhesive is deposited to form a layer of the object.

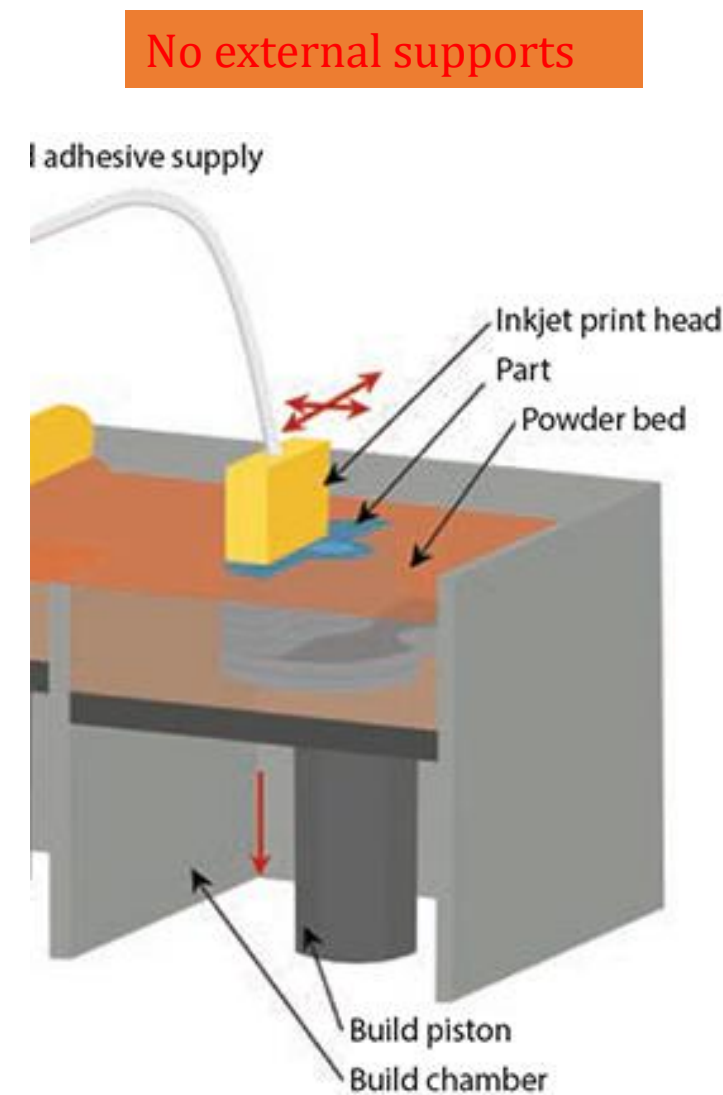


Once a layer is completed, the fabrication **piston moves down** by the thickness of a layer, and the process is repeated until the entire object is formed within the powder bed.

After completion, the object is elevated and the extra powder **brushed away** leaving the object.

No external supports are required during fabrication since the powder bed supports overhangs.

After building up the whole part, a heat treatment is applied by **curing** at 120°C for 2 h.



No external supports

piston moves down

Features

Three-dimensional printing offers the advantages of **speedy fabrication** and low materials cost.

In fact, it's probably the **fastest** of all RP methods.

Recently **color output** has also become available.

However, there are **limitations** on resolution, surface finish, part fragility, and available materials.

The closest competitor to this process is probably fused deposition modeling (FDM).



Advantages

(1) High speed.

Fastest 3D printer to date. Each layer is printed in seconds, reducing the prototyping time of a hand-held part to 1 to 2 hours.

(2) **Versatile**. Parts are currently used for the automotive, packaging, education, footwear, medical, aerospace and telecommunications industries. Parts are used in every step of the design process for communication, design review and limited functional testing.



(3) **Simple to operate.**

The office compatible Zcorp system is straightforward to operate and **does not require** a **designated technician** to build a part. The system is based on the standard, off the shelf components developed for the ink-jet printer industry, resulting in a reliable and dependable 3D printer.

(4) **No wastage of materials.** Powder that is not printed during the cycle can be reused.

(5) **Color.** Enables complex color schemes in RP-ed parts from a full 24-bit palette of colors.

Disadvantages

(1) Limited functional parts.

Relative to the SLS, parts built are much **weaker**, thereby limiting the functional testing capabilities.

(2) Limited materials.

The materials available are only **starch** and **plaster-based materials**, with the added option to infiltrate wax using the ZW4 Waxer.

(3) Poor surface finish.

Parts built by 3D printing have a relatively poorer surface finish and post-processing is frequently required.

Applications

- Leading athletic **shoe companies**, such as **Adidas**, have used this RP system to radically reduce prototype development time and communicate in new ways
- to produce computer-sculpted models to **assist artists** with complex modeling. Parts produced find applications with **computer gamers**, **animators**, pre-Hollywood mock-ups designers, and sculpturing artists.

